

2x 500MW FLOATING PHOTOVOLTAIC PLANT BATHYMETRIC AND TOPOGRAPHICAL SURVEY DRAFT REPORT



Location: Gache-Gache Bay, Nyenje, Kariba, Zimbabwe
Owner \ Client: Green Hybrid Power (Pvt) Ltd
Owner's Engineer: LTI-Reenergy (Germany)
Technical Consultants: China Energy (International) assisted by iX Engineers (RSA)
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1 Table of Contents

1.1	Project Overview	4
1.2	Purpose of the Survey	5
2	Project Background	5
2.1	Overview of Lake Kariba and Site Significance	5
2.2	Need for Floating Solar	6
2.3	Importance of Accurate Data	6
3	Objectives of the Survey	7
4	Study Area Description	8
4.1	Location and Accessibility	8
4.2	Reservoir Characteristics (Water Levels)	8
4.3	Environmental Conditions	8
5	Survey Methodology	9
5.1	Bathymetric Survey Method	9
5.1.1	Equipment and Boat Setup	9
5.1.2	Sound Velocity and Depth Principles	9
6	Topographical Survey Method	10
7	Control Survey	10
7.1	Datum and Coordinate System	10
7.2	Benchmark Establishment and Control Network	10
8	Equipment and Software Used	11
8.1	Equipment	11
8.2	Software Used	13
9	Results and Analysis	13
9.1	Bathymetric Results	13
9.1.1	Depth ranges	13
9.1.2	Reservoir bed morphology	13
9.1.3	Identification of steep slopes, sedimentation zones and obstructions	13
9.2	Topographical Results	14
9.2.1	Terrain features	14
9.2.2	Shoreline definition	14
9.2.3	Elevation ranges	14

9.3	Combined Surface Model	14
9.3.1	Digital Terrain Model (DTM)	14
9.3.2	Contours and Cross Sections	14
9.3.3	Integration of land and water datasets	14
10	Suitability Assessment for Floating Solar	14
10.1	Identification of suitable zones	14
10.2	Site Suitability Constraints and Engineering Implications	14
11	Challenges Encountered	17
12	Limited Access to Shoreline Areas	17
13	11.3 Wildlife Encounters and Safety Risks	18
14	11.4 Operational Implications	18
14.1	Dynamic Water Level Fluctuations	18
14.1.1	Challenge	19
14.2	Adverse Weather Conditions	19
14.2.1	Challenge	19
14.3	Data Gaps and Accessibility Issues	19
14.3.1	Challenge	20
14.4	Equipment and Environmental Conditions	20
14.4.1	Challenge	20

TABLE OF FIGURES

Figure 1: Shows a 1 in 50000 map highlighting the geographical location of the project area- Gache-Gache Bay, Nyenje, Kariba	5
Figure 2: Highlights 1000-hectares area coverage for the floating solar plant	7
Figure 3: Anchor Design Layout	8
Figure 4: Highlights bathymetric lines spaced at 30m intervals	11
Figure 5: Hi-Target v300 GNSS	14
Figure 6: Hi-Target HD550 Single Beam Echo Sounder	14
Figure 7: The DJI Matrice 350 RTK (M350) unmanned aerial vehicle (UAV) integrated with the Zenmuse L2 LiDAR sensor, utilized for high precision shoreline mapping and digital terrain modelling	15
Figure 8: Highlights depth of the 1000 hectares area of interest	17

Figure 9: Highlights Reservoir Bed Morphology of Gache-Gache Bay, Nyenje, Kariba	18
Figure 10: Digital Terrain Model of Gache-Gache Bay, Nyenje, Kariba	19
Figure 11: Digital Terrain Model Slope Map of Gache-Gache Bay, Nyenje, Kariba	20
Figure 12: Contours and cross-section map of Gache-Gache Bay, Nyenje, Kariba	21
Figure 13: Suitability map for siting floating solar plant-Gache -Gache Bay, Nyenje, Kariba	23
Figure 14: Locality Map showing areas with tree hazards	25

1. Introduction

1.1 Project Overview

The proposed **2 × 500 MW Floating Solar Development at the Gache-Gache bay of the Nyenje area of Lake Kariba** represents a significant advancement in Zimbabwe's renewable energy strategy. The project aims to harness solar energy on the expansive water surface of Lake Kariba, creating a hybrid energy system that complements existing hydroelectric generation.

Located at **Gache-Gache Bay, Nyenje, Kariba**, the project benefits from a relatively sheltered reservoir environment, making it suitable for large-scale floating photovoltaic (FPV) deployment. The integration of floating solar technology offers key advantages, including reduced land use conflict, improved panel efficiency due to water cooling effects, fisheries industry enhancement and reduced evaporation losses from the reservoir.

However, successful implementation requires a detailed understanding of the reservoir's bathymetry, shoreline characteristics, and environmental dynamics, including wind-driven waves and seasonal water level fluctuations.

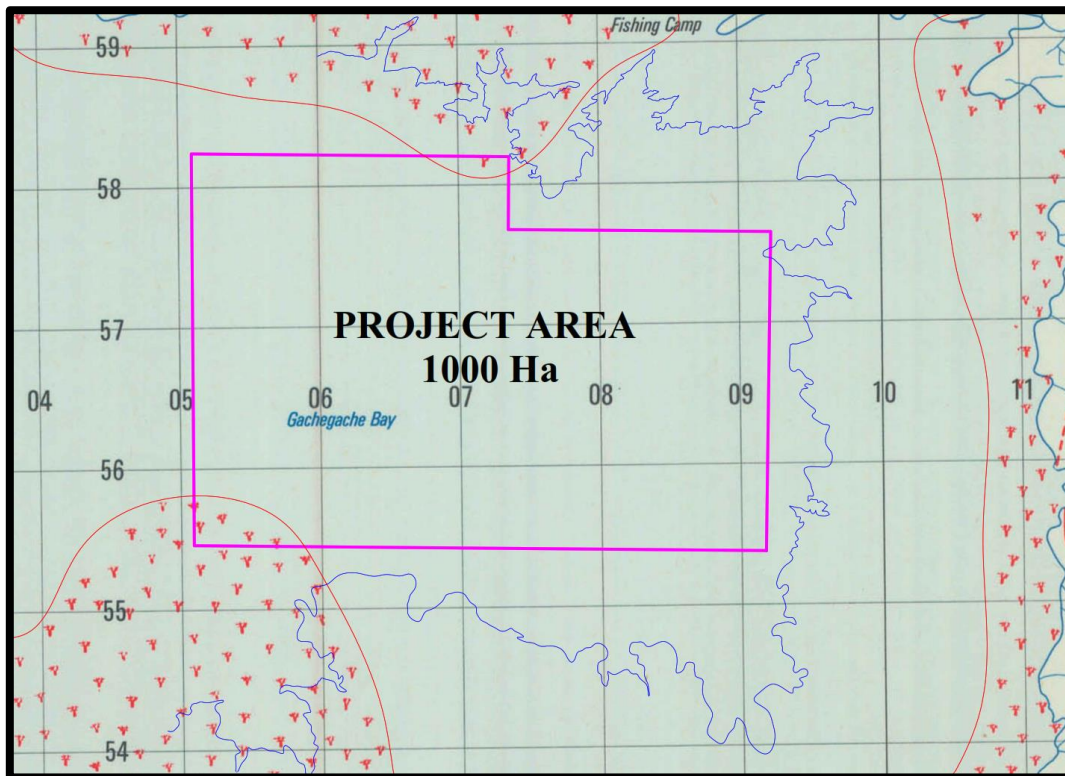


Figure 1: Shows a 1 in 50000 map highlighting the geographical location of the project area- Gache-Gache Bay, Nyenje, Kariba

1.2 Purpose of the Survey

The purpose of this survey was to generate **high-resolution bathymetric and topographical data** to support the design and implementation of the floating solar infrastructure.

Commissioned by the client \ owner, **GHP [Green Hybrid Power (Pvt) Ltd]** through their International Technical Consultants, China Energy and under the supervision of regional Consultants **iX Engineers**, the local Key Informatics and Geodetic team executed the survey in order to provide critical spatial data required for:

- Anchoring and mooring system design
- Floating array layout optimization
- Cable routing and infrastructure planning

- Development of a Digital Terrain Model (DTM)
- Development of other areas linked to the project like road network; construction of amenities like workshops, accommodation, and environmental considerations

Whist GHP plans to execute the project in two phases of 500MW, covering approximately two by 500 hectares of water surface area, the execution team used its discretion and found it beneficial to survey and report on the entire 1000 hectares as presented in this report and shown in figure 1 above.

2 Project Background

2.1 Overview of Lake Kariba and Site Significance

Lake Kariba, situated on the Zambezi River between Zambia and Zimbabwe, is a titan of African infrastructure, boasting a total storage capacity of approximately 181 cubic kilometers and a surface area exceeding 5,400 square kilometers. The dam wall stands 128 meters high, holding back a basin that is critical to regional energy security. The proposed 2 x 500 MW Floating Solar System is specifically situated within the Mashonaland West Province of Zimbabwe, at coordinates 16°39'10.29" S, 28°56'45.52" E. While the total project area encompasses 1,000 hectares or 10 square kilometers (less than 0.2% of Lake surface area), the high-resolution bathymetric survey focused on 4000 hectares, covering deep-water zones and the transitional shoreline near the Gache-Gache/Father-Girl fishing camp in the Nyenje area. This specific site is significant due to its proximity to existing transmission nodes and its relatively sheltered bathymetric profile, which reduces the mechanical stress on the floating platform's mooring lines.

The project site at **Gache-Gache Bay** is strategically positioned due to:

- Proximity to existing power infrastructure (about 14km to 330kV main Grid)
- Relatively stable bathymetric conditions
- Reduced exposure to extreme wave action



Figure 2: Highlights 1000-hectares area coverage for the floating solar plant

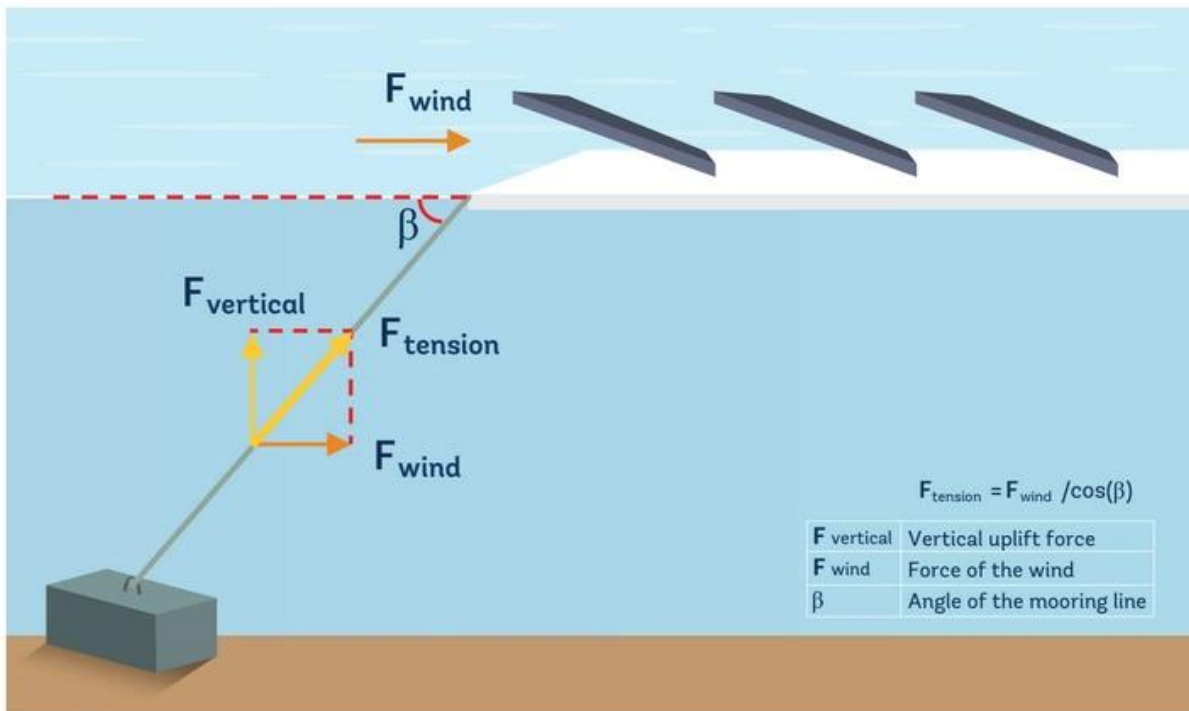
2.2 Need for Floating Solar

Zimbabwe currently faces persistent electricity shortages that hamper industrial growth and daily life. The integration of a floating solar farm at Kariba offers a dual-benefit solution: it provides immediate renewable energy to the national grid while preserving valuable terrestrial land for agriculture or conservation. Furthermore, the floating panels reduce water evaporation from the reservoir surface, effectively "saving" water for hydroelectric generation during peak demand periods.

2.3 Importance of Accurate Data

Precise bathymetric and terrain data are non-negotiable for:

- Anchor Design: Determining the penetration depth and type of anchors based on bed slope.



Source: Adapted from SERIS.

Note: Larger tilt angles (β) will lead to higher tension for the same amount of wind load.

Figure 3: Anchor Design Layout

- Mooring Systems: Calculating the scope and tension of lines to accommodate water level changes.
- Platform Layout: Avoiding underwater obstructions and optimizing sun exposure.
- Cable Routing: Mapping the safest path for submerged cables to reach the shore-based substation.

3 Objectives of the Survey

- 1 To determine reservoir bed levels: To produce high-accuracy bathymetric maps of the underwater topography.
- 2 To map shoreline and adjacent land: Conduct a topographical survey to facilitate the transition from water-based to land-based infrastructure.

- 3 To identify suitable zones: Pinpoint areas with stable slopes and adequate depths for PV installation.
- 4 To support engineering design: Provide the XYZ datasets necessary for the modeling of anchors, float arrays, and electrical routing.

4 Study Area Description

4.1 Location and Accessibility

The project site is located at **Gache-Gache Bay, Nyenje, Kariba**, within Mashonaland West Province.

Access is primarily via boat from Kariba town, with travel distances of approximately **23–25 km**, depending on launch point

4.2 Reservoir Characteristics (Water Levels)

Lake Kariba experiences significant annual and multi-year water level fluctuations driven by upstream rainfall in the Zambezi basin. Over the past five years, the lake has seen variations of up to 5 meters between the peak of the rainy season and the end of the dry season

Lake Kariba Annual Water Level Variations

Table 1: Highlights Lake Kariba Annual Water Level Variations as per Zambezi River Authority

4.3 Environmental Conditions

The site is subject to prevailing North-Easterly winds. According to Zambezi River Authority data, wind speeds can average 15–20 km/h, with gusts during storms creating significant

wave chop. Surface currents remain generally low (< 0.5 m/s) but can increase near the main river channel.

5 Survey Methodology

5.1 Bathymetric Survey Method

5.1.1 Equipment and Boat Setup

The bathymetric data was acquired using a Hi-Target V550 Single Beam Echo Sounder integrated with a Hi-Target V300 dual-frequency GNSS receiver. The sensors were mounted on a 2m bathymetry pole secured to a 10m x 5m vessel (boat). Survey lines were spaced at 30m intervals to ensure comprehensive coverage. The boat maintained an average velocity of 7 km/h, ensuring a stable platform that resisted wave-induced oscillations and provided high-density sounding points.



Figure 4: Highlights bathymetric lines spaced at 30m intervals

5.1.2 Sound Velocity and Depth Principles

Sound velocity is a critical variable in underwater acoustics, influenced by water temperature, salinity, and pressure. To ensure accuracy, the Echo Sounder utilized the Time-of-Flight (ToF) principle. The depth (D) was calculated using the formula:

$$D = \frac{1}{2} \cdot V \cdot t + k$$

Where:

v= Velocity of sound in water (approx. 1480 m/s at the site).

t= Two-way travel time of the acoustic pulse.

k = Transducer draft (the depth of the sensor below the water line).

The single beam Echo sounder system transmitted high-frequency pulses (200 kHz) to the reservoir bed, and the reflected signal processed to determine the precise vertical distance to the bed.

6 Topographical Survey Method

A hybrid approach combining ground-based GNSS and UAV photogrammetry was employed.

- I. GNSS RTK: A Hi-Target v300 receiver was used for ground control. The V300 offers a horizontal accuracy of $\pm (8\text{mm} + 1\text{ppm})$ and a vertical accuracy of $\pm (15\text{mm} + 1\text{ppm})$.
- II. UAV LiDAR: A DJI Matrice 350 RTK equipped with a LiDAR L2 Sensor payload was utilized to map the shoreline and rugged terrain near the Gategate fishing camp, providing high-density point clouds even in vegetated areas.
- III. Ground Control Points (GCPs): GCPs were coordinated with 20+ iterations per point to ensure sub-centimeter georeferencing accuracy for the aerial datasets.

7 Control Survey

7.1 Datum and Coordinate System

To ensure seamless integration between the bathymetric and topographical datasets, all survey observations were referenced to the World Geodetic System 1984 (WGS 84) datum. The data was projected using the Universal Transverse Mercator (UTM) Zone 35 South coordinate system. This unified spatial framework is essential for the precise alignment of the floating PV arrays with the terrestrial anchoring points and the existing power grid infrastructure.

Ground Control Points and Control Points

Table 2: Established Ground Control Points and Control Points

Name	N	E	Z
CP 1	8158662.38	705934.05	485.497
CP 2	8158646.20	705920.30	485.320
CP 3	8158657.45	705935.34	485.309
CP 4	8157943.95	707768.15	478.340
CP 5	8158721.84	710399.79	484.279
CP 6	8155760.37	709682.69	477.531
DRTK 1	8154511.66	708802.50	477.385
DRTK 3	8155062.69	706967.37	477.032
DRTK 6	8158654.42	705930.63	485.659
DRTK 6A	8158657.45	705935.34	485.309
DRTK PEG 7	8158540.04	709740.74	477.991

Benchmark Establishment and Control Network

A robust control network was established to provide the geodetic foundation for the entire project area. Primary control points were coordinated using a Hi-Target V300 dual-frequency GNSS receiver operating in RTK mode. To achieve sub-centimeter horizontal and vertical precision, each key benchmark was occupied for a minimum of 20 iterations (epochs), ensuring the mitigation of multipath errors and atmospheric interference.

A total of eight (8) permanent control points were established and strategically distributed across the site to facilitate both the UAV shoreline mapping and the bathymetric vessel's positioning. The final adjusted coordinates and elevations for these benchmarks are detailed in Table 3 below.

8 Equipment and Software Used

8.1 Equipment

- I. GNSS: Hi-Target V300 (Dual-frequency, Multi-constellation).



Figure 5: Hi-Target v300 GNSS

- II. Echo Sounder: Hi-Target HD550 Single Beam Echo Sounder.

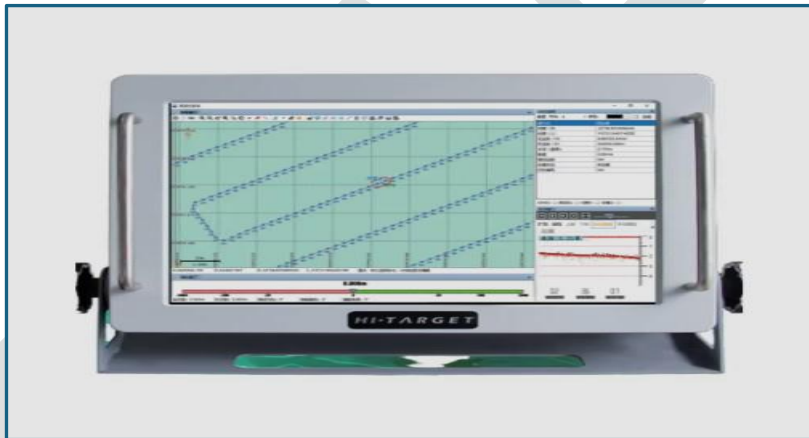


Figure 6: Hi-Target HD550 Single Beam Echo Sounder

- III. UAV: DJI Matrice 350 RTK Drone paired with the Zenmuse L2 LiDAR Sensor

The precision and accuracy of the DJI Matrice 350 RTK, paired with the Zenmuse L2 LiDAR sensor, are defined by the integration of the aircraft's RTK positioning, a high-performance IMU, and the LiDAR module's own ranging capabilities, as detailed in Table 3 below.

Summary Table

Parameter	Specification (at 150 m AGL)
Horizontal System Accuracy	5 cm
Vertical System Accuracy	4 cm
Lidar Ranging Precision	2 cm
RTK Horizontal Positioning	1 cm + 1 ppm
RTK Vertical Positioning	1.5 cm + 1 ppm
Max Returns	5 Returns (240,000-point cloud pts/s captured in a single return)

Table 3: DJI Matrice 350 RTK Drone paired with the Zenmuse L2 LiDAR Sensor Parameters



Figure 7: The DJI Matrice 350 RTK (M350) unmanned aerial vehicle (UAV) integrated with the Zenmuse L2 LiDAR sensor, utilized for high precision shoreline mapping and digital terrain modelling

8.2 Software Used

- I. Hi-Max Hydrographic Processing Software: For hydrographic data acquisition.
- II. DJI Terra: For LiDAR point cloud processing.
- III. Civil 3D: For DTM generation and contouring.
- IV. Agisoft Metashape: For DTM and Ortho-Mosaics generation

9 Results and Analysis

9.1 Bathymetric Results

The bathymetric survey results indicate that the project area contains **in excess of 800 hectares of water surface** with depths ranging between **2 m and 10 m**, which is considered highly suitable for floating solar development.

This depth range falls within the optimal operational window for reservoir-based floating photovoltaic (FPV) systems, where:

- Anchoring systems can be efficiently designed and installed
- Mooring line lengths remain manageable and cost-effective
- Structural stability of floating platforms is enhanced
- Maintenance and installation activities are simplified

The identified area therefore presents a **significant and viable zone for large-scale deployment** of the proposed **2 × 500 MW floating solar arrays**, with minimal geotechnical and operational constraints.

9.1.1 Depth ranges

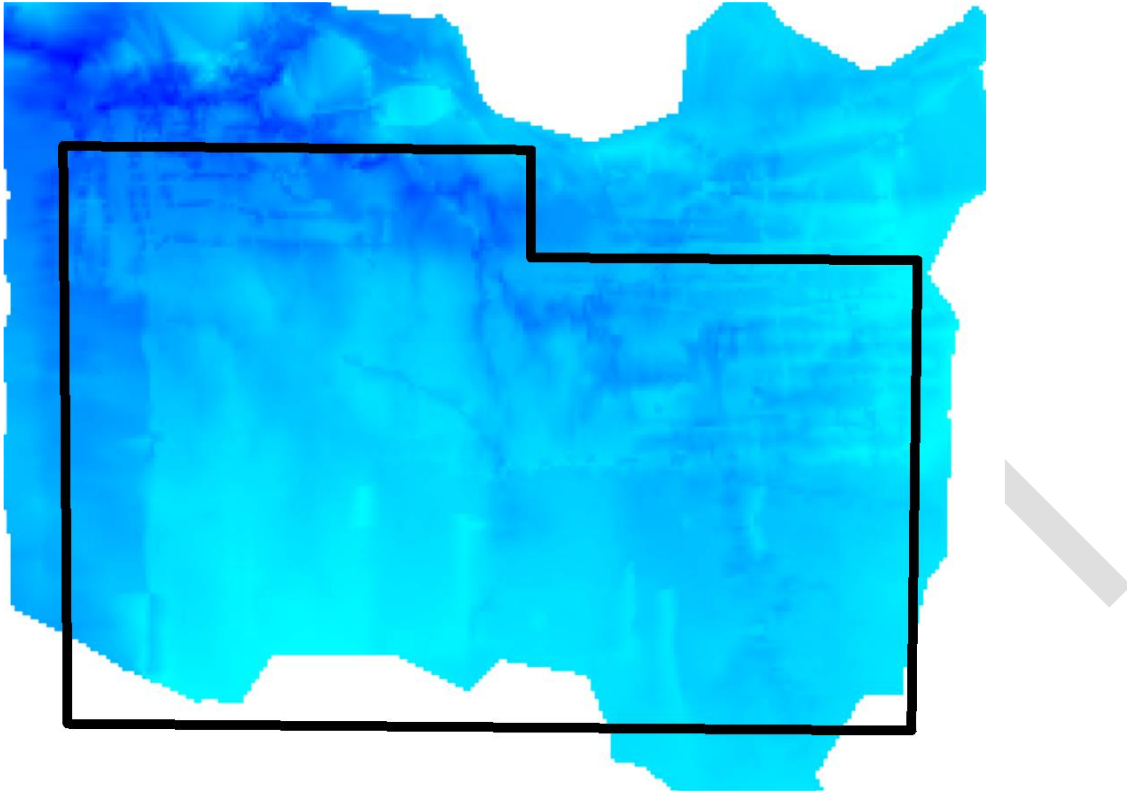


Figure 8: Highlights depth of the 1000 hectares area of interest

9.1.2 Reservoir bed morphology

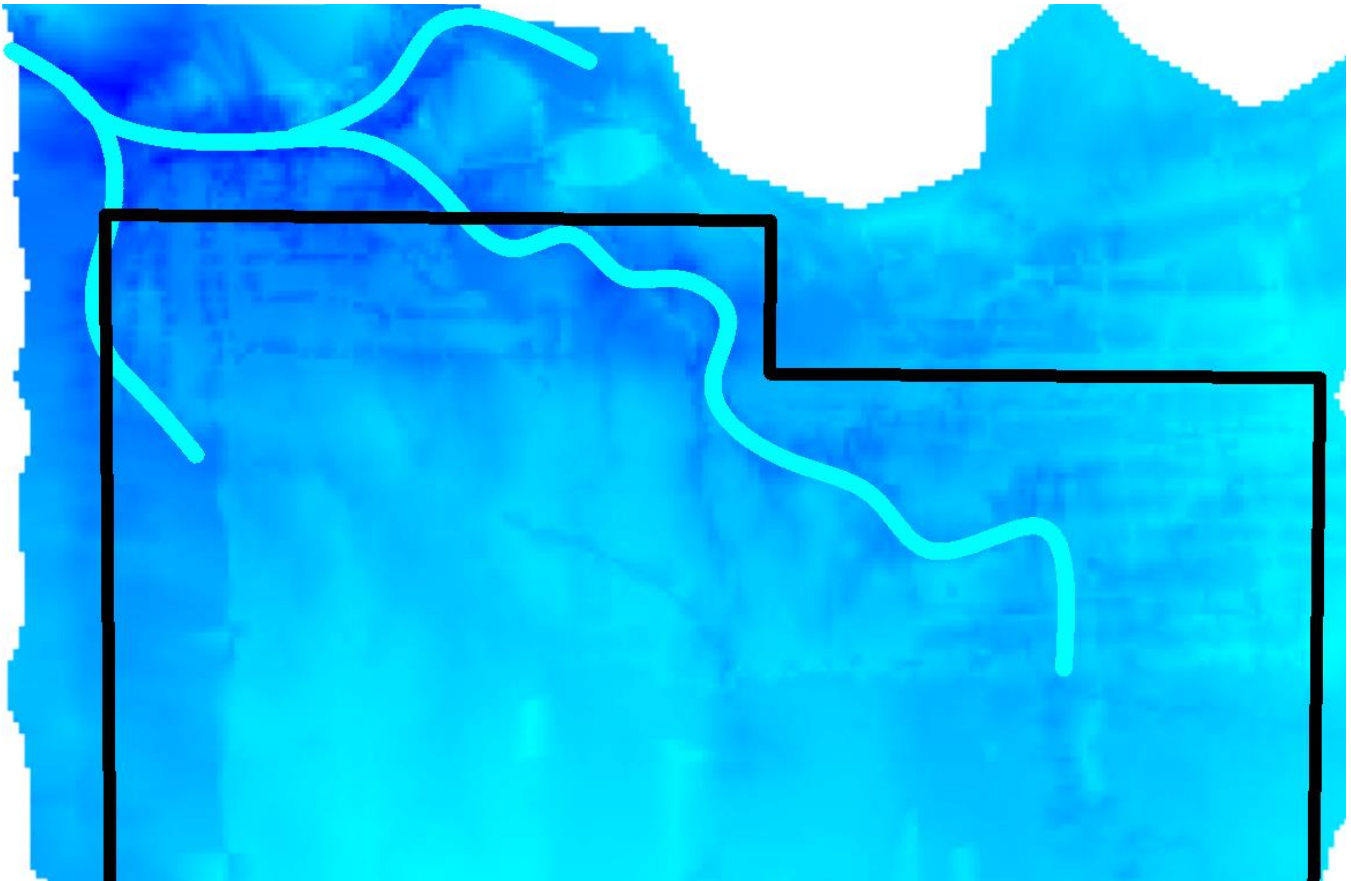


Figure 9: Highlights Reservoir Bed Morphology of Gahe-Gache Bay, Nyenje, Kariba

9.2 Topographical Results

Terrain features

Shoreline definition

Elevation ranges

Combined Surface Model

Digital Terrain Model (DTM)

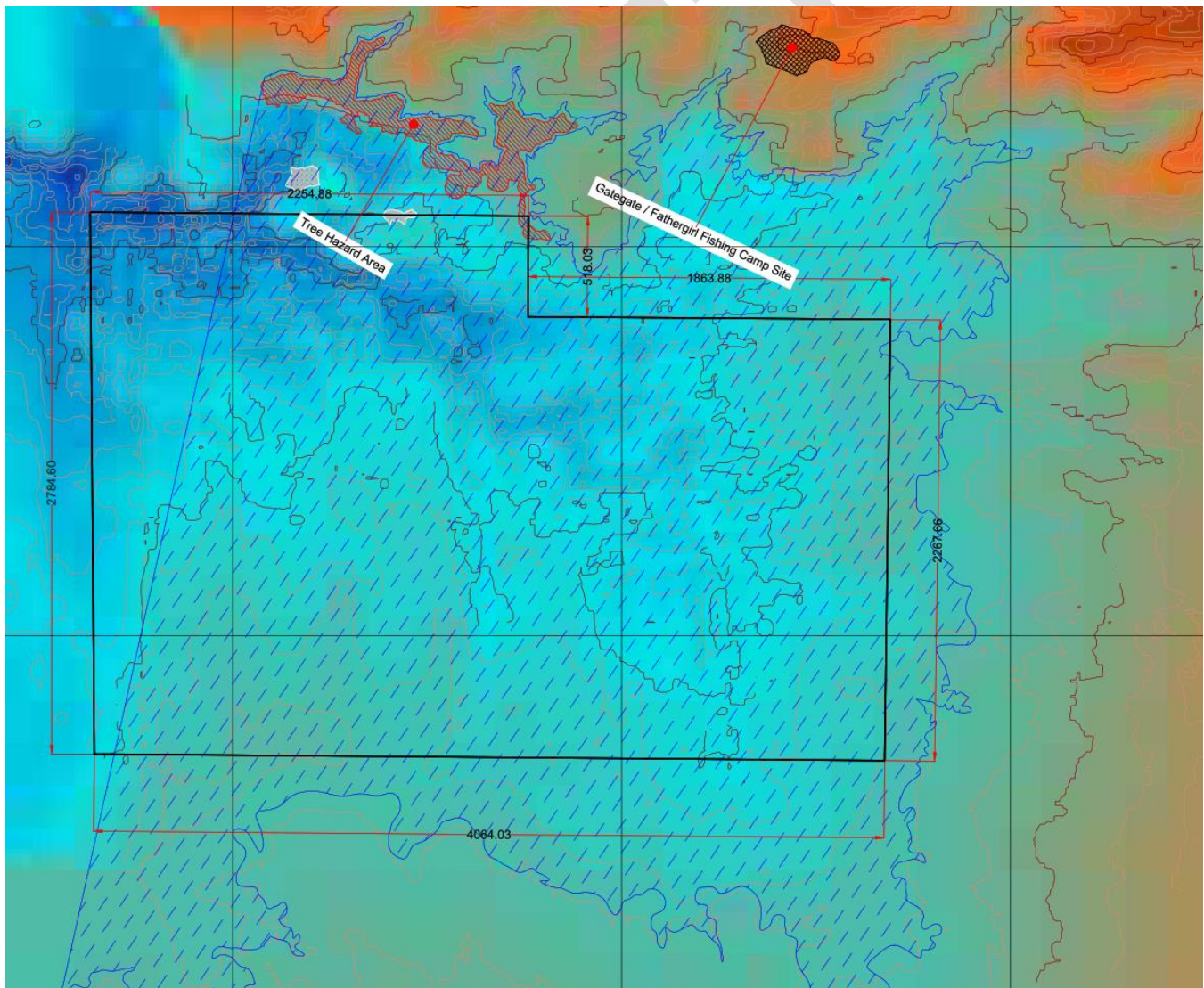


Figure 10: Digital Terrain Model of Gache-Gache Bay, Nyenje, Kariba

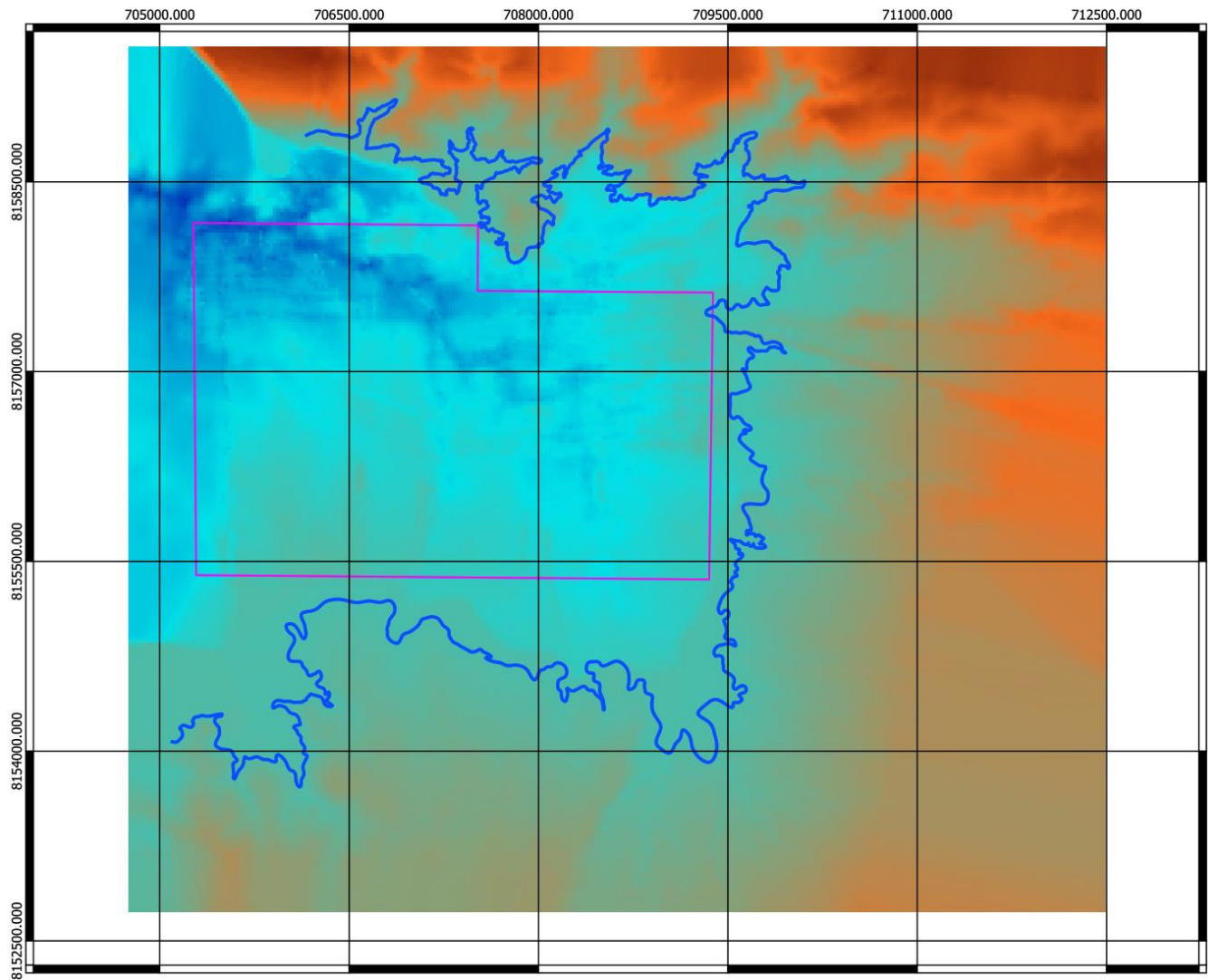


Figure 11: Digital Terrain Model Slope Map of Gache-Gache Bay, Nyenje, Kariba

9.2.1 Contours and Cross Sections

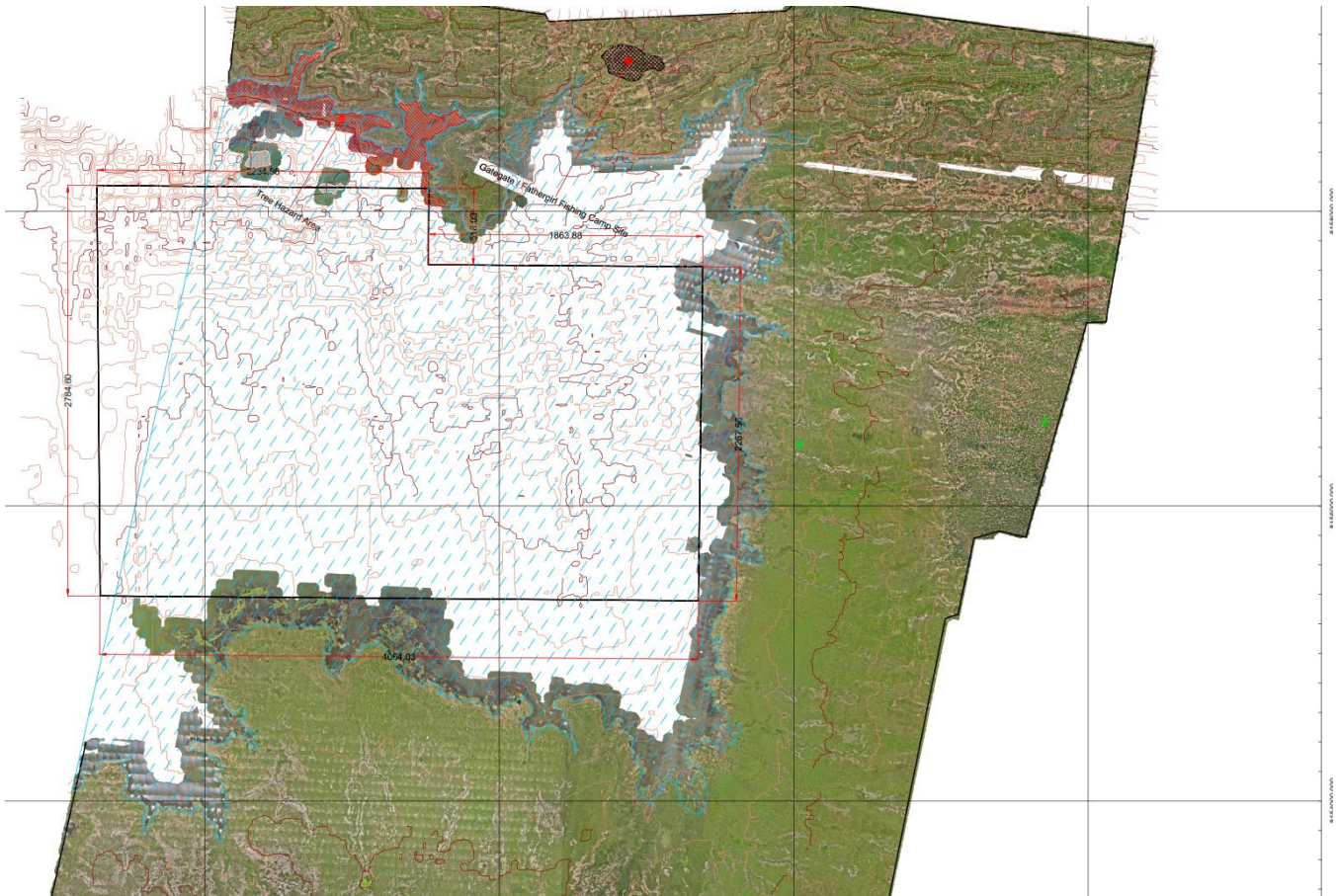


Figure 12:Contours and Cross-Section map of Gache-Gache Bay, Nyenje, Kariba

10 Suitability Assessment for Floating Solar

Identification of suitable zones

- Depth (Anchoring feasibility)
- Slope stability

10.1 Site Suitability Constraints and Engineering Implications

The feasibility of the proposed floating solar array is governed by several physical and operational constraints. Identification of these zones was critical for the longevity of the mooring system and the safety of the floating infrastructure.

1 Shallow water bathymetry

A practical engineering rule is:

- Less than about 1 m depth: generally unsuitable for utility FPV, because very shallow water creates problems for flotation, seasonal grounding, weed growth, and anchoring; one recent technical-potential study excluded areas below 1 m entirely.
- About 2–10 m depth: often a good working range for many reservoir-based projects, especially where water levels do not fluctuate excessively. In this range, bottom anchoring is still feasible and construction/maintenance are simpler than in very deep water. This is an engineering inference based on guidance favoring moderate depths and common bottom-anchored layouts.
- Around 10–30 m depth: still workable, but mooring design becomes more important, and costs can rise depending on bed conditions, slope, and drawdown.
- Very deep water: can still be used, but it is usually less economical unless the site has strong reasons to justify it, because mooring lines get longer and design must account for larger movements and load effects.

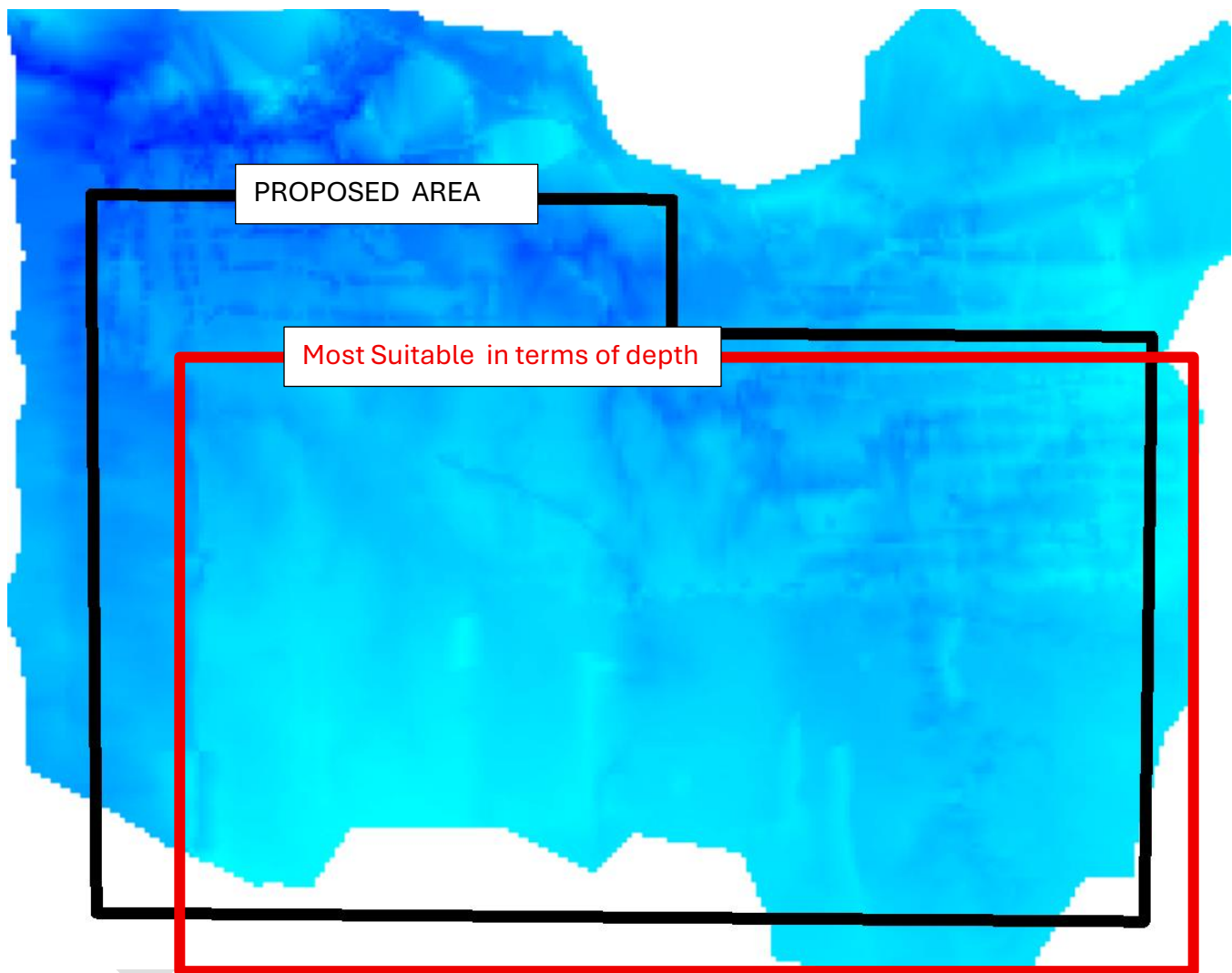


Figure 13: Suitability map for siting floating solar plant-Gache -Gache Bay, Nyenje, Kariba

2 Localized Island feature

Effect: The presence of an island at the specified coordinates ([N], [E]) creates a physical obstruction that dictates the spatial layout of the array. Furthermore, islands can cause "wind tunneling" or shadowing effects, which reduce the overall energy yield of the nearby PV strings.

Engineering Impact: The design must maintain a professional buffer zone around the island to prevent mooring line entanglement and to allow for natural water circulation.

3 Submerged dead trees

Effect: The proposed site contains extensive stands of submerged leadwood trees, which present a primary geomorphological constraint for anchoring operations. If an anchor is deployed within a cluster of these underwater hazards, it may fail to achieve the required design tensile strength. Furthermore, submerged timber poses a significant risk of snagging mooring lines during seasonal water level fluctuations, potentially compromising the structural integrity of the floating array.

Engineering Impact: A Zenmuse L2 LiDAR sensor was deployed to capture high-resolution aerial datasets and multispectral imagery, facilitating the precise georeferencing of submerged and semi-submerged dead timber. By determining the exact spatial distribution of these obstructions, the survey allows for the strategic selection of specialized anchor types such as helical or gravity-based systems where traditional drag-embedment anchors would be compromised by the dense leadwood stands.

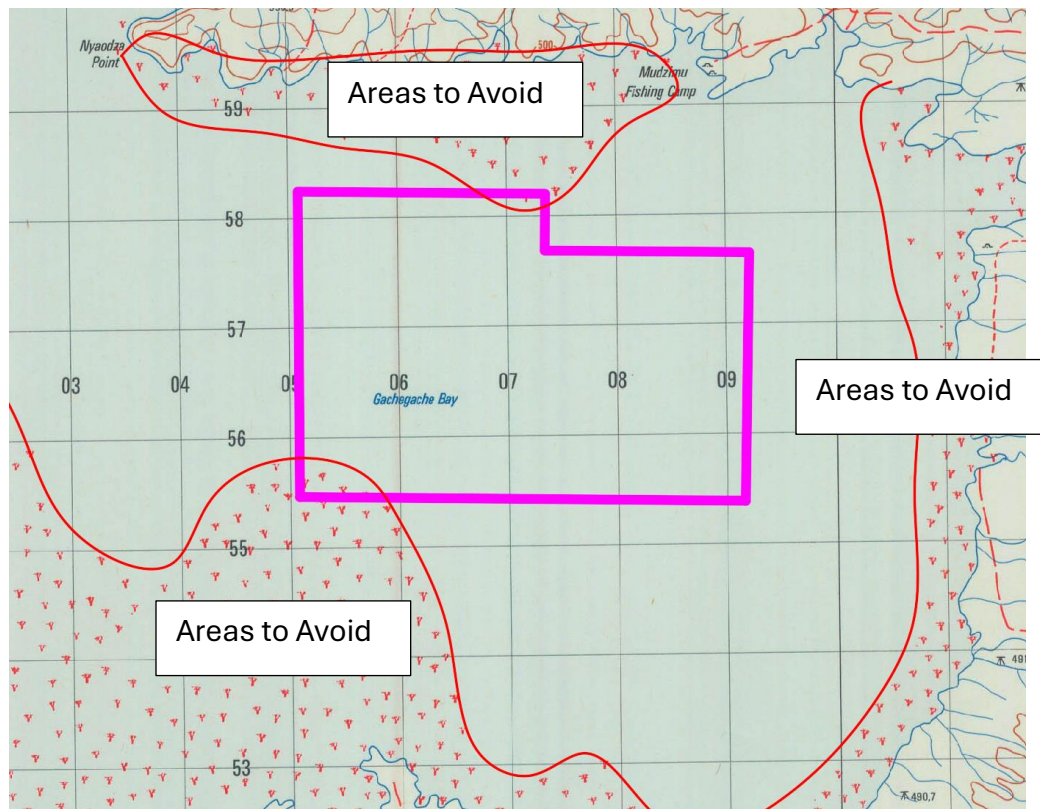


Figure 14: Locality Map showing areas with tree hazards

4 Steep bathymetric drop-offs

Effect: Areas with rapid changes in elevation are unstable for traditional anchoring. Steep slopes can lead to "anchor creep," where the mooring system gradually slides down the slope under the tension of wind and wave loads, potentially causing a catastrophic failure of the array's positioning.

Engineering Impact: Slope analysis was conducted to facilitate the installation limit to zones with a gradient of less than 10%, ensuring long-term geotechnical stability.

11 Challenges Encountered

12 Limited Access to Shoreline Areas

- Due to dense vegetation, shallow depths, and submerged obstacles, the survey vessel could not access large portions of the reservoir margins.
- This resulted in:
 - Reduced data coverage along the perimeter of the dam basin
 - Need for interpolation of data in inaccessible zones
- These limitations may affect the accuracy of volume calculations and detailed edge profiling.

13 11.3 Wildlife Encounters and Safety Risks

- The survey team encountered dangerous wildlife both on land and within the reservoir, including:

Hippopotamuses (hippos)

Crocodiles

- These animals posed significant safety risks, particularly during:

Boat operations in the water

Shoreline access and equipment setup

- As a precaution:

Certain areas were avoided or surveyed from a safe distance

Survey operations were conducted with heightened vigilance and reduced working time in high-risk zones

14 11.4 Operational Implications

These challenges collectively resulted in:

- Increased survey duration
- Reduced spatial coverage in critical near-shore zones
- Necessity for cautious navigation and conservative data acquisition approaches

14.1 Dynamic Water Level Fluctuations

Lake Kariba is subjected to significant stage-height variations driven by Zambezi River Authority (ZRA) dam operations and seasonal inflow.

14.1.1 Challenge

During the survey period, fluctuating water levels meant that the "shoreline" was a moving target. Data collected on different days required precise vertical synchronization to a stable datum.

Mitigation

All bathymetric soundings were corrected in post-processing using daily water level readings from the Zambezi River Authority, ensuring all XYZ data points were reduced to the WGS 84 / UTM Zone 35S vertical reference.

14.2 Adverse Weather Conditions

The expansive fetch of Lake Kariba allows for the development of significant wave heights, particularly during "Guti" weather patterns characterized by strong North-Easterly winds.

Challenge

High wave amplitudes introduced vertical oscillations into the boat's movement, which at some point distorted echo sounder readings. Additionally, wind speeds exceeding 10 m/s posed a risk to UAV stability during LiDAR acquisition.

Mitigation

Bathymetric operations were restricted to early morning windows when the lake surface was most quiescent. For the UAV survey, the DJI Matrice 350 RTK's advanced wind resistance (up to 10 m/s) was utilized, with flights paused during peak gust periods to maintain IMU accuracy.

14.3 Data Gaps and Accessibility Issues

The project site's location, 23–25 km from Andora Harbour and shorelines created logistical hurdles.

Challenge

Certain, shallow-water transition zones of depths < 1.5m were inaccessible by the survey boat due to the risk of propeller damage and hull grounding.

Mitigation

These data gaps were bridged using the DJI M350 RTK LiDAR during low-water periods. By overflying the shoreline at high resolution, we achieved a seamless topography to bathymetry model, filling the gaps where the vessel could not navigate.

14.4 Equipment and Environmental Conditions

Challenge

The presence of steep shoreline cliffs interfered with GNSS signal reception.

Mitigation

The DJI M350 RTK LiDAR system facilitated a comprehensive aerial survey of the project site, generating high-fidelity topographic datasets used to validate and cross-reference the bathymetric bed levels. To ensure maximum positional accuracy and minimize RTK signal latency or baseline interference, the GNSS base stations were strategically relocated closer to the bathymetric survey area, maintaining a short baseline for robust real-time corrections.

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